

10. Confirming and Reporting Findings

Freeze the selected test, measure its effect on reserved evidence and assess uncertainty, materiality and scope.

A finding needs more than a high discovery score

EFFECT

How much worse is the selected region or condition?

Report the metric and comparison.

EVIDENCE

How much independent support is available?

Assess uncertainty and selection.

RELEVANCE

How many cases are affected?

Which operational decisions and populations does it cover?

Keep severity, affected share, uncertainty and materiality together.

Freeze the test before confirmation

PREDICTIVE FINDING

Predictor and leaf representation.
Medoids and selected region rule.
Conditions, loss and support checks.
Hypothesis and reporting threshold.

AGENTIC FINDING

Task family and presentation rule.
Conversation constructor if used.
Reference and source-access policy.
System and scoring versions.

Reserve confirmation outcomes from all candidate and structural selection decisions.

Evaluate the frozen membership rule on new cases

DISCOVERY

Find and select a region on discovery outcomes.

Store its medoid vectors and assignment rule.

Record why that region was selected.

CONFIRMATION

Assign new cases using the frozen rule.

Compute regional and complement loss.

Estimate contrast, share and uncertainty.

A fresh confirmation sample tests whether the selected weakness persists under the declared data-generating scope.

Confirmed one-shot AML region

Measurement	Recorded confirmation value
Region / sample count	12,575 / 200,000
Affected share	6.2875%
Region / rest mean Brier	0.091412 / 0.015888
Brier contrast	0.075524
Simultaneous 95% interval	[0.069631, 0.081417]

The confirmed excess loss is positive for this frozen region and synthetic population.

Match uncertainty to the observation unit

CONTINUOUS LOSS CONTRAST

Recompute region and complement summaries together.

Use a justified influence-function or resampling method.

Preserve shared-case dependence.

BINARY FAILURE RATE

Declare exactly what counts as failure.

Use a binomial model when independent cases share a probability.

Handle clustered variants jointly.

Resampling estimates uncertainty from the available evidence; it does not add independent cases.

Why a 100% observed failure rate can be uncertain

Illustration: the claim is that the failure probability exceeds a predeclared 50% threshold.

Observed failures	Rate	One-sided 95% lower bound
2 / 2	100%	22.4%
5 / 5	100%	54.9%
10 / 10	100%	74.1%

For n independent failures out of n, the exact lower bound is $L = 0.05^{(1/n)}$.

State the hypothesis and meaningful threshold

EXAMPLE HYPOTHESES

Null: failure probability is at most 50%.

Alternative: it exceeds 50%.

The group and event are fixed first.

FIVE FAILURES OUT OF FIVE

At the null boundary, probability = $0.5^5 = 0.03125$.

This is the one-sided p-value for this outcome.

A p-value measures incompatibility with the null model; the effect and its practical meaning remain separate.

Declare the family of confirmation tests

WHY THE FAMILY MATTERS

With 20 independent null tests at 5%,
the expected false positives are 1.

Chance of at least one is about 64%.

CHOOSE THE ERROR TARGET

Specify tests, levels and conditions used
for the reporting decision.

Choose family-wise or false-discovery
control under its assumptions.

Fresh confirmation addresses adaptive discovery; multiplicity control addresses the declared set of confirmation claims.

Interpret a false-discovery rate target

WHAT IS CONTROLLED

False-discovery proportion = false findings / selected findings.

Use zero when none are selected.

FDR is its expectation across repeated analyses.

WHEN BH APPLIES

Start with valid p-values.

Check independence or suitable positive dependence.

Declare the family and target q before selection.

At $q = 5\%$, the target concerns the expected false fraction, not a fixed quota for an individual report.

Apply Benjamini–Hochberg to valid p-values

Illustrative family of five tests. FDR target $q = 5\%$.

Rank / test	Raw p	BH cutoff	Selected
1 / A	0.001	0.010	Yes
2 / B	0.012	0.020	Yes
3 / C	0.032	0.030	No
4 / D	0.200	0.040	No
5 / E	0.600	0.050	No

At $q = 5\%$, the largest passing rank is 2; select every test through that rank.

Interpret factor effects within the tested design

MEASURED ASSOCIATION

Estimate differences across levels and conditions.

Retain base-case dependence and relevant interactions.

Report effect sizes and uncertainty.

SUPPORTED CONCLUSION

State the tested tasks, population and allocation.

Identify sparse or unscorable cells.

Separate association from a causal mechanism.

A significant factor does not by itself identify the worst level or explain why the system failed.

Assess weakness and repair separately

WEAKNESS FINDING

A frozen region has elevated loss on reserved evidence.

Report severity, prevalence, support and scope.

REPAIR FINDING

A frozen proposed repair improves a declared comparison with the baseline.

Check targeted regions and overall performance.

Measure any repair benefit and adverse tradeoffs on reserved cases under the same assessment plan.

Deliver a finding with its evidence limits

CLAIM

Metric, contrast, threshold and affected population.

Frozen rule and conditions.

SUPPORT

Counts, estimates, intervals and test-family decision.

Sources and reproducible records.

SCOPE

Materiality and required action.

Unresolved evidence and confirmation limits.

Separate any repair claim.

Next: generate the report from these records and check both its statements and its required coverage.